

Announcements

- Assignment 1
 - due Thu 24 Sep
 - for completion

Today

- Continuous time, birth-death IBM

Consistent themes

- Scale transition
 - lower level processes $\rightarrow$ higher level outcomes
- Emergence in the scale transition
 - given rules operating at one level, what behavior or dynamics emerges at the next?
- Algorithms to math abstraction
 - distributions are abstractions always in algorithms
 - sometimes we get an abstraction (named distribution or ODE) for a population-level outcome but not necessarily (i.e. algorithm is primary)

Simulate multiple event types

r_i : birth, death, immigration, predation, mutation ...

Algorithm (Gillespie)

while condition

 intensity = $r_1 + r_2 + \dots + r_n$

 draw waiting time w from $\text{Exp}(\text{intensity})$

$t = t + w$

 decide which event occurred, $p_i = r_i / \text{intensity}$

 update state (do event, rates, etc)

Multicategory stochastic events

- Generalization to binary stochastic events and Bernoulli distribution
- Categorical distribution
 - `multicategory_stochastic_events.md`
- Use for Gillespie with multiple events

Birth-death IBM

r_i : birth_ind₁ ... birth_ind_N, death_ind₁ ... death_ind_N

no change in principle:

Algorithm (Gillespie)

while condition

intensity = $r_1 + r_2 + \dots + r_n$

draw waiting time w from $\text{Exp}(\text{intensity})$

$t = t + w$

decide which event occurred, $p_i = r_i / \text{intensity}$

update state (do event, rates, etc)

Birth-death IBM

r_i : birth_ind₁ ... birth_ind_N, death_ind₁ ... death_ind_N

minor equivalent variant to decide which event:

Algorithm (Gillespie)

while condition

 intensity = $r_1 + r_2 + \dots + r_n$

 draw waiting time w from $\text{Exp}(\text{intensity})$

$t = t + w$

 decide which event type occurred, birth/death

 decide which individual it happened to

 update state (do event, rates, etc)